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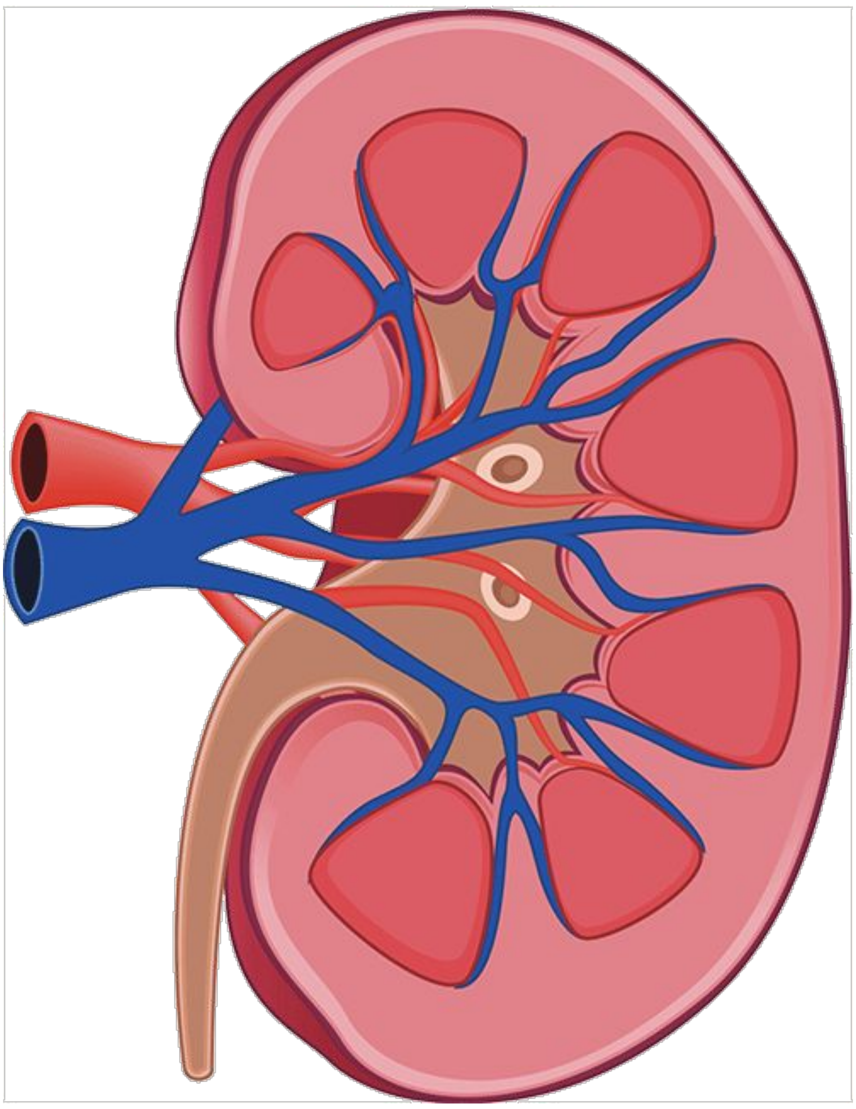
GEMS

VERSION 1.1

Nephrology



Goodpasture Syndrome

Facts

It is a combination of:

- Acutely progressive glomerulonephritis
- Pulmonary alveolar haemorrhage

It is an autoimmune disease

Features

- Haematuria
- Hemoptysis
- Impaired renal function test

Investigations

1. Blood test
   → Anti-glomerular basement membrane antibodies (anti-GBM antibodies)
2. Kidney biopsy
   → Shows crescentic glomerulonephritis
3. Chest X-ray / CT scan
   → Shows patchy interstitial infiltration (intra-pulmonary bleeding)
4. Lung biopsy there are any lung involvement

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Goodpasture Syndrome & Differential Diagnosis

Condition	Features	Investigation
Alport syndrome	• Haematuria • Hemoptysis • Abnormal U&E • Loss of sight • Loss of hearing	
α-1 Antitrypsin deficiency	• Hemoptysis • Jaundice	
Churg Strauss (Eosinophilic granulomatosis with polyangiitis)	• Asthma • Eosinophilia	p-ANCA
Wegener's granulomatosis (Granulomatosis with polyangiitis)	• Haematuria • Nasal septum perforation • Epistaxis	c-ANCA
Hemolytic uremic syndrome	• Haematuria • Blood diarrhoea	

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Itching

Chronic Renal Failure

- Itching → ↑ Serum urea
- Tiredness → ↓Erythropoietin → Anaemia
- Peripheral oedema
- Hyperpigmentation

Liver Failure

- Itching → ↑ Serum urea
- Ascites
- Jaundice
- Bleeding → ↓ Platelet production

Polycythemia rubra vera

- Itching
- Red skin → ↑ Haemoglobin
- Splenomegaly
- Burning sensation in fingers and toes
- Gout

Scabies

- Itching
- Line tracks on skin (burrows)

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Rhabdomyolysis

Facts

It is a condition of **dying of skeletal muscles** and as a result releasing:

- **Myoglobin**
- **Potassium**
- **Creatinine kinase**

Common scenarios

- Trapped for several hours under heavy object
- Fall followed by lying for long period of time on floor
- An elderly with frequent fall after acute kidney injury
- IV drug user lying on floor not moving for long
- Severe exertion or dehydration e.g. marathon
- Severe crash injury

Features

- Haematuria
- Hypotension
- AKI
- ↑ Creatine kinase

Summary

1. Prolonged immobilisation
   → Muscle ischaemia
   → Rhabdomyolysis
2. Myoglobin
   → Red colour due to haem

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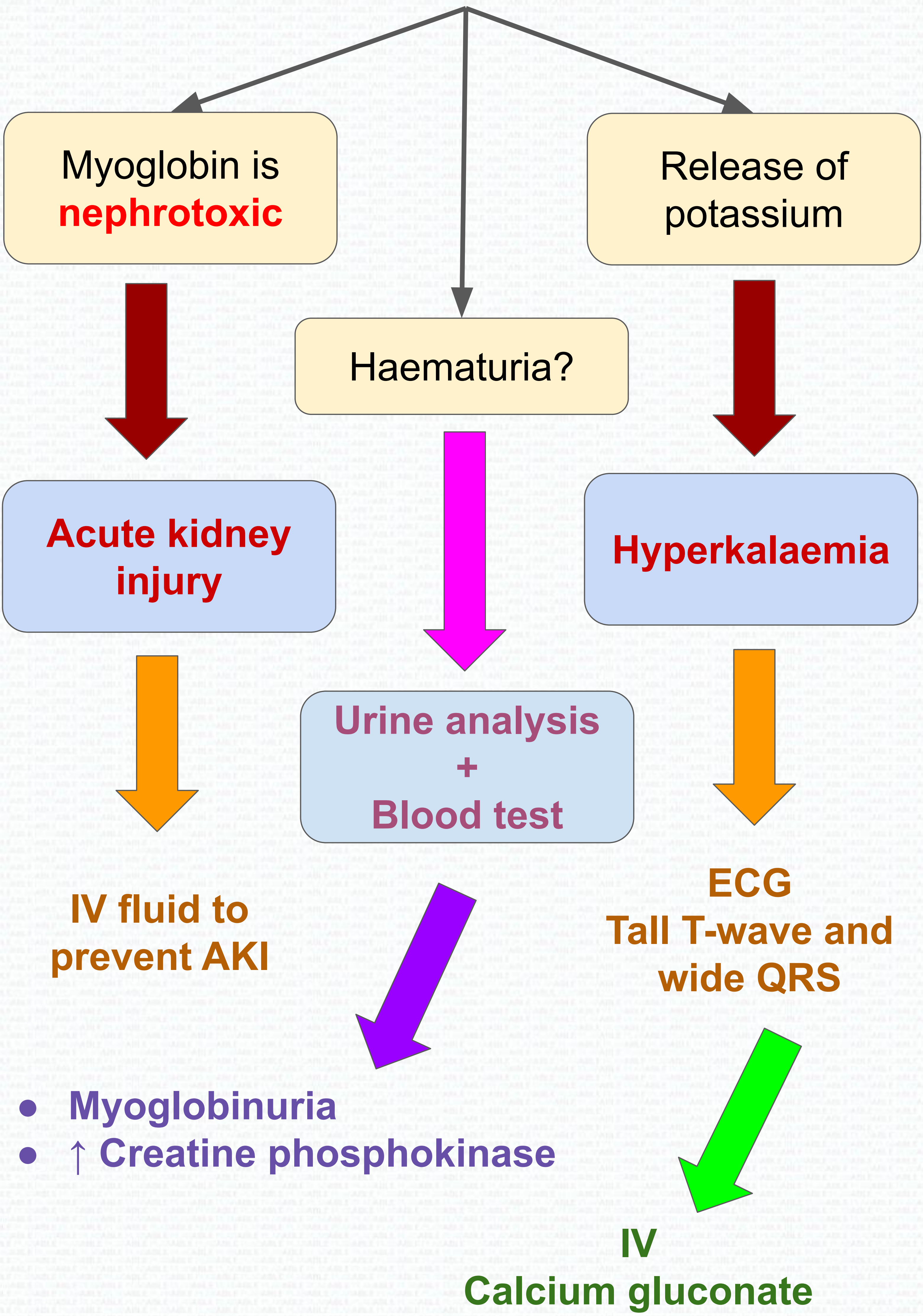
Rhabdomyolysis Management



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Acute Tubular Necrosis

Causes

- Massive haemorrhage
  → Hypotensive shock
  → High creatinine
- Prolonged renal ischaemia
  → Reduced renal perfusion
  → Tubule necrosis

Acute tubular necrosis is the commonest renal cause of AKI

Management

- Maintain hydration and perfusion

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Interstitial Nephritis

Features:

- Allergy
  → e.g drug
  → Causing rash or fever reaction
- Haematuria

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Hyponatraemia

Brain trainer:

An man is being treated for sepsis with ATB and IV. fluids. A short time later he has hyponatraemia. What is the most likely cause ?

➔ **Iatrogenic**

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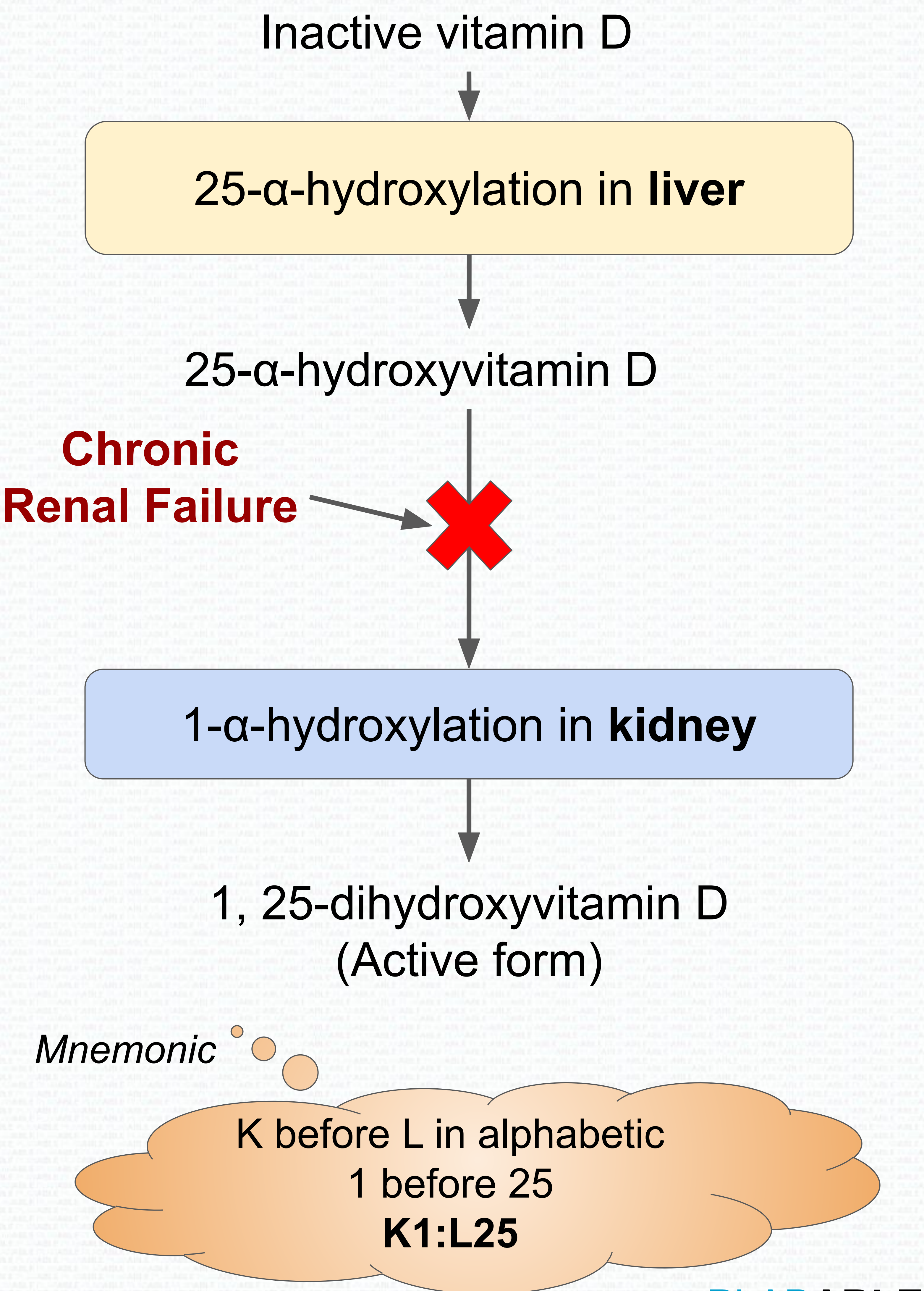
Vitamin D Deficiency



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Pyelonephritis

Iinflammation of **pelvis** of the kidney

Contributing factors

Pregnancy, stone, vesico-ureteric reflux, diabetes

Common cause

- Ascending UTI

Symptoms

- Dysuria
- Frequency
- Urgency
- Lower abdominal pain
- Fever
- Loin or back pain
  (Costovertebral angle tenderness)

Lower UTI

Pyelonephritis

ACUTE

Sudden development:

- Fever
- Rigors
- Loin pain
- Repeated UTI

CHRONIC

- Hypertension
- Repeated UTI
  → Renal scarring
- No active infection

IMPORTANT

Small Kidney + Hypertension
= Bilateral renal artery stenosis
= Or chronic pyelonephritis
ACE-inhibitor is contraindicated!

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Management Of Pyelonephritis

Investigation

- Urinalysis shows blood, protein, nitrate, leukocyte esterase
- Urine culture and sensitivity **before** commencing antibiotics

In acute pyelonephritis:
Start empirical antibiotics immediately once sample has been sent.
E.Coli* = *Most common cause of UTI

Treatment

Acute pyelonephritis	
• Non-pregnant woman • Men • Patient with indwelling catheters	• Cefalexin 500mg BD to QDS depends on severity • Or ciprofloxacin 500mg BD
Children	1st line: cefalexin 2nd line: co-amoxiclav only if sensitive
Pregnancy *who does not require hospital admission	Cefalexin 500mg BD to TDS ***If admitted then give IV cefuroxime**

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Proteinuria

Facts

Protein (trace up to +1) in dipstick test

- Can be normal
- Can happen in patient who exercise e.g. gym attendees

If **no symptoms** and **healthy**

→ Repeat the test

If still high

→ 24hrs urine specimen test

→ Protein creatinine ratio

DO NOT refer to renal clinic because of an 'isolated proteinuria'

Further investigation before a referral is always required!

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Nephrotic Syndrome

In children

- Peak between 2-5 years old
- 80% are due to minimal change glomerulonephritis

Features

1. Proteinuria (> 3g/ 24hr)
2. Hypoalbuminemia (< 30g/L)
3. Oedema
4. Hyperlipidaemia
5. Hypercoagulable state - loss of antithrombin III
6. Predisposition to infection - loss of immunoglobulins

Investigations

1. 24hrs urinary protein specimen
2. Blood test - Albumin level
3. Renal biopsy - definitive diagnostic test

Foamy or frothy urine = High protein

Treatment

- High dose oral steroid

→ 90% have good prognosis

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Common Cause Of Nephrotic Syndrome

In Children	
Minimal change of glomerulonephritis	
In Adults	
• Caucasians (White) • Unspecified ethnicity	• Afro-caribbeans (Black) • American • Hispanics
Membranous glomerulonephritis	**Focal segmental glomerulonephritis**
Others	
• Infection • Malignancy • Rheumatoid drugs	

In PLAB1, most common cause for nephrotic syndrome in adult > 40 years old is **membranous glomerulonephritis**

Prognosis

⅓ → Remission
⅓ → Partial Remission
⅓ → Progress to end stage renal failure

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Minimal Change Glomerulonephritis

Minimal change diseases always account for nephrotic syndrome

Features

- Children
- Periorbital oedema, oedema
- Nephrotic syndrome
- Normotensive
- Selective proteinuria
  → Albumin and transferrin
- Renal biopsy
  → Electron microscopy shows **fusion of podocytes**

Example

6 years old boy presented with progressively:

- Swelling of face, scrotum and legs
- Urine is frothy
- Fusion of podocytes shows on electron microscopy

RED are all the hints!

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Minimal Change Disease

Brain trainer:

A boy presents with significant proteinuria. What should you check before referring this boy to nephrology ?

➔ **Serum albumin levels**

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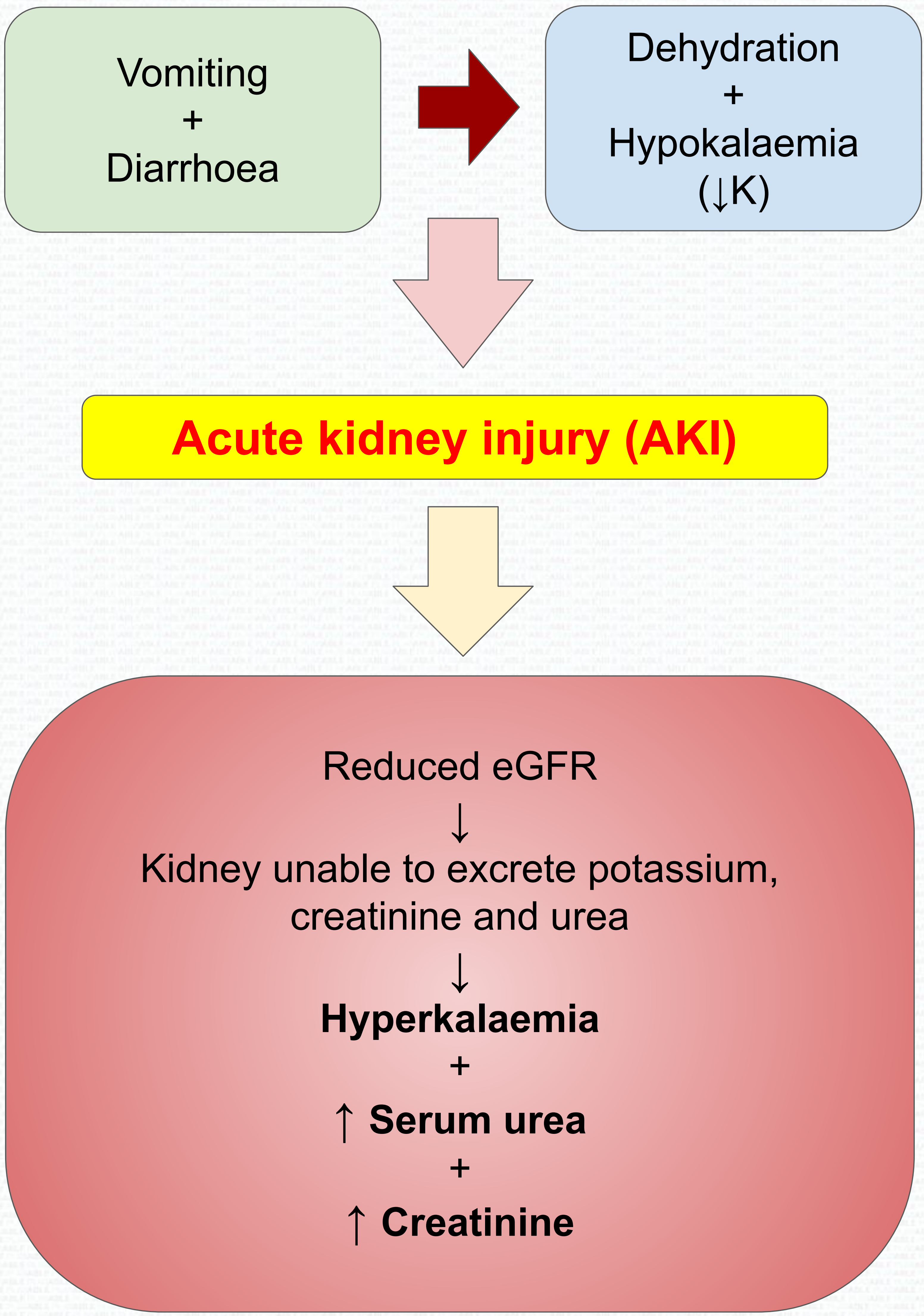
Hypovolaemia



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Contrast Induced Nephropathy

To **reduce risk** of contrast in CT scan or any contrast study

- Drink plenty of fluid
- Stop nephrotoxic drugs e.g. Metformin
- IV fluid with normal saline (0.9% NaCl)
  - Pre- and post-op
  - Particularly in high risk patient
  - e.g. Elderly with diabetes

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Blood Gas Abnormality

1

Acidaemic

pH < 7.35

Alkalaemia

pH > 7.45

2

Respiratory acidosis
$PaCO_2$ > 6.0 kPa

(Respiratory compensation for metabolic alkalosis)

Respiratory alkalosis
$PaCO_2$ < 4.7 kPa

(Respiratory compensation for metabolic acidosis)

3

Metabolic acidosis
Bicarbonate
< 22 mmol/L

(Renal compensation for respiratory alkalosis)

Metabolic alkalosis
Bicarbonate
> 26 mmol/L

(Renal compensation for respiratory acidosis)

CO_2 is an acid. Bicarbonate is an alkali

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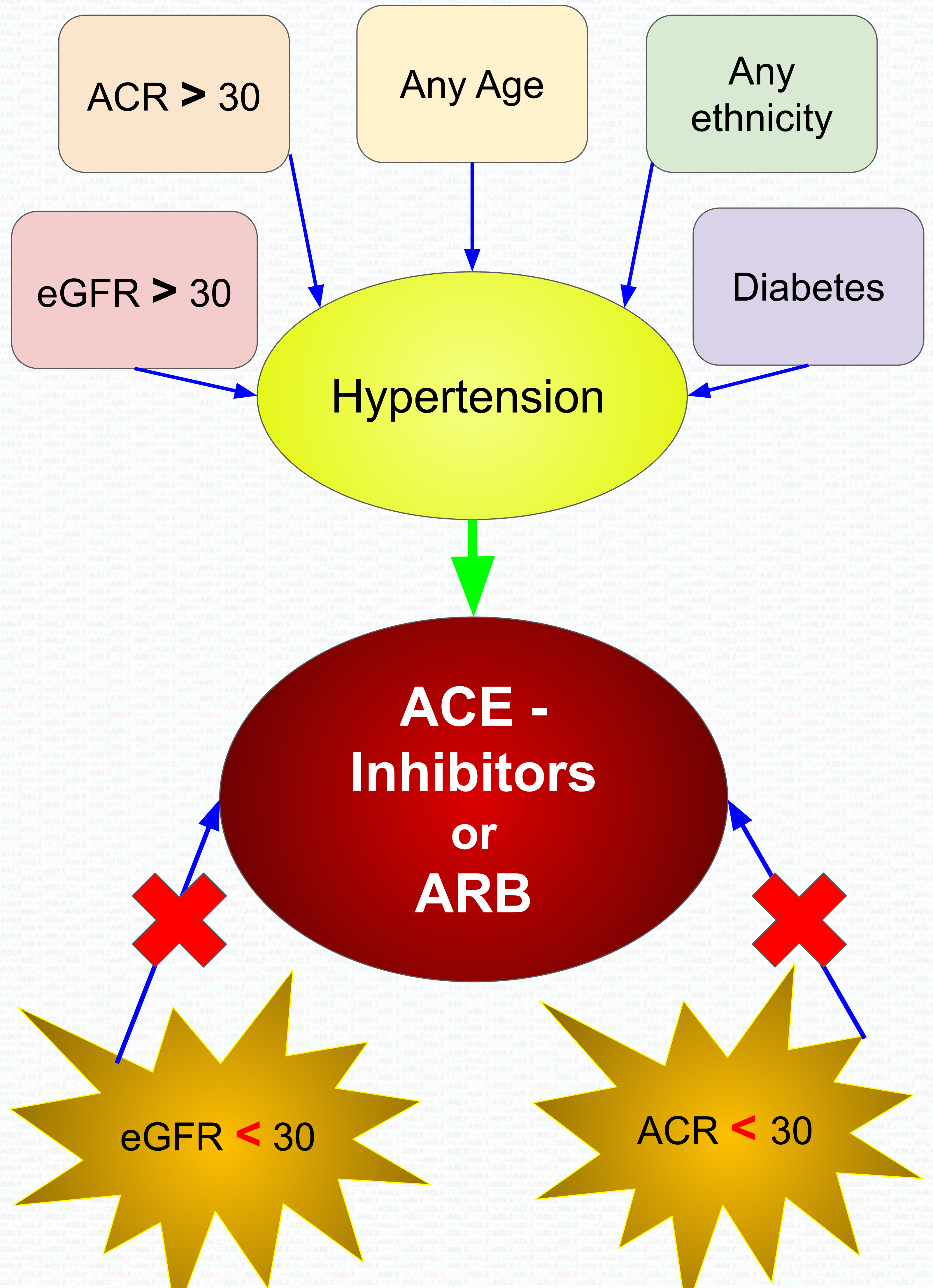
Hypertensive In Chronic Kidney Disease



ACR: Albumin Creatinine Ratio

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Types Of Glomerulonephritis

Presented with nephritic syndrome	**Presented with nephrotic syndrome**
• **Haematuria** • **Hypertension**	• **Proteinuria** • **Oedema**
Rapid progressive glomerulonephritis (Crescentic glomerulonephritis)	**Minimal change diseases**
• Rapid onset • Often presented as AKI • Caused by Goodpasture → Haematuria → Hemoptysis Or ANCA +ve vasculitis	• Accounted for 80% of nephrotic syndrome • Good response to steroid • Fusion of podocytes show in electron microscopy via renal biopsy • Mostly idiopathic or Hodgkin's or steroid induced

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Types of glomerulonephritis

Presented with nephritic syndrome	**Presented with nephrotic syndrome**
• **Haematuria** • **Hypertension**	• **Proteinuria** • **Oedema**
IgA nephropathy (Berger's diseases)	**Membranous glomerulonephritis**
• Young adult with haematuria 1-2 days after URTI	• Presentation: → Proteinuria → Nephrotic syndrome → CKD • Idiopathic or caused by infection, rheumatoid drugs or malignancy • ⅓ resolved • ⅓ response to cytotoxics • ⅓ develops to CKD

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Types of glomerulonephritis

Presented with nephritic syndrome • **Haematuria** • **Hypertension**	**Presented with nephrotic syndrome** • **Proteinuria** • **Oedema**
Mesangioproliferative glomerulonephritis • Young adult with haematuria 1-2 days after URTI	**Focal segmental glomerulosclerosis** • Idiopathic or secondary to HIV/heroin • Presentation: → Proteinuria → Nephrotic syndrome → CKD

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When To Suspect Chronic Kidney Disease?

Anaemia	Hypocalcaemia
Hyperphosphatemia	Small Kidney on ultrasound < 9 cm

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Hypocalcaemia

Symptoms

- Tingling
- Numbness
- Paresthesia
- Involuntary spasm/cramps

CKD can cause

1,25 dihydroxyvitamin D3 deficiency

→ ↓ Calcium absorption

→ Hypocalcemia

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Autosomal Dominant Polycystic Kidney Diseases

Features

- Haematuria
- Hypertension
- Loin or flank pain

Often associated with intracranial aneurysm

Investigation

- Ultrasound kidney, ureters and bladders

Autosomal dominant

= 50% of offspring (1st generation) will be affected

Can lead to progressive CKD

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Haemolytic Uremic Syndrome (HUS)

In children

1. Eating undercooked contaminated food
2. E.Coli → Produce verotoxin
3. Profuse **diarrhoea**
4. **Bloody** diarrhoea
5. After 2-14 days
   → **Uremia (Acute renal failure)**
6. + Features of anaemia e.g fatigue, pallor

Features

1. Haemolytic anaemia (Haemolysis)
2. Uremia (Acute renal failure)
   → Haematuria
   → Proteinuria
   → ↑ Urea and creatinine
3. Thrombocytopenia (Low platelets)

Treatment

- IV Fluid
- ± Blood transfusion
- ± Dialysis
- Plasma exchange (in very severe case only)

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Haemolytic Uremic Syndrome (HUS)

NEVER give antibiotics as it releases more toxin by killing *E.Coli*

Suspect **thrombotic thrombocytopenic purpura** if they have **features 1, 2, 3, fever and neurological manifestation**

Diseases	*Features*	*Investigation*
Polycystic Kidney Diseases (ADPKD)	• Haematuria • Hypertension	Ultrasound
Goodpasture Syndrome	• Haematuria • Hemoptysis	Anti-GBM antibodies
Wegener's (Granulomatosis with Polyangiitis)	• Haematuria • Hemoptysis • Nasal/Sinus symptoms	c-ANCA
Haemolytic Uremic Syndrome (HUS)	• Haematuria • Bloody Diarrhoea	

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Haematuria After A Upper Respiratory Tract Infection

IgA glomerulonephritis	Post-streptococcal glomerulonephritis
1-2 **days** after URTI	1-2 **weeks** after URTI
Main Features: • Haematuria	Main Features: • Proteinuria
Patient factor: • Young males	Patient factor: • ↓ Complement (C3) Level • Humps on electron microscopy in renal biopsy

Causative organism
→ Group A beta-haemolytic Streptococci (*Streptococcus Pyogenes*)

Treatment
→ Mainly supporting as it is self-limiting

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Abnormal Kidney Size Causes

Large Kidney

- ADPKD
  → Multiple cysts makes kidney bigger
- Obstructive uropathy
  → Due to stone or enlarged prostate
  → Hydronephrosis
  → ↑ Size

Small Kidney

- Chronic pyelonephritis
- Chronic glomerulonephritis
- Hypertensive renal diseases
  → Small and scarred kidney
- Bilateral renal artery stenosis

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Abnormal Kidney Size Causes

Examples:

Patient with hypertension with elevated creatinine, u/s show small kidney

→ Hypertensive renal diseases

Patient with enlarged prostate presented with prostatism (urinary frequency, post-void dripping, hesitancy)

→ Obstructive uropathy + Hydronephrosis

→ Ultrasound will show large kidney

Patient with dilated pelvicalyceal system and is young

→ Reflux nephropathy

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Haemodialysis

Indications:

Persistent high potassium	Acidosis e.g. Hyperventilation
Pulmonary oedema, pericarditis	Fluid overload with anuria

Example:

Patient had renal transplant and presented with above features?

Causes:
Transplant rejection or Host-versus-graft:

❌ Treat hyperkalemia
❌ Give IV calcium gluconate followed by insulin and glucose

∵These **DO NOT** treat all the above features as kidney does not work, potassium and toxin will accumulate again.
Extra fluid can worsen fluid overload problem

BEST treatment: Hemodialysis

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